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Experimental study on the shear behavior of fully grouted bolts

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HIGHLIGHTS

- The deflecting shape, bending angle and failure characteristic of the bolts were described and analyzed in detail.
- The effect of bolting angles and grout strengths on the bolt contributions was determined quantitatively.
- Bolt deflecting behavior near the rock joint was clearly reveal by the strain monitoring.
- The founding refines the understanding of the failure mode of sheared bolts in weaker rock.

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ABSTRACT

Failures of fully grouted bolts due to the combination of tensile and shear loads are commonly observed, but the bolting mechanism is not clearly revealed. Direct shear tests were conducted on three groups of specimens with different bolting angles or grout strengths, and shear loads, shear displacements and bolt strains were measured. The deflecting shape, the bending angle and the failure characteristic of the bolts were described in detail. The relationship between the shear load and the shear displacement was analyzed, and the internal forces were determined according to the bolt strains or the shear displacement of the joint. The test results show that the bolting angle and grout strength play significant roles in influencing the bolt contribution. With increasing the bolting angle, the bolt contribution rises first and then reduces, and they obey a parabola relationship. Particularly, the contribution of the bolt is less than its ultimate tensile load by a large margin when the bolt steeply intersects the rock joint. A higher shear contribution will be mobilized in a weaker grout. It is also observed that the failure of the bolt occurs at the bolt-joint intersection rather than at the two plastic hinges.

1. Introduction

Fully grouted bolts were continually used as a main support tool of rock mass stabilization in civil and mining engineering more than a century. In the past, fully grouted bolts were usually characterized as pure tension elements to support the unstable rock block, and the bolting mechanism and their performance under traction were deeply studied [1–5]. However, more and more failure cases of fully grouted bolts and field monitoring results indicate that the shear movement of rock joints will arouse a considerable shear force in the bolt and that the shear force together with the axial force causes the failure of the bolt [6–10]. It has been progressively realized that traction bolting theories fail to reveal the anchoring mechanism of fully grouted bolts subjected to tensile and shear loads and hence, further investigations on this subject need to be carried out for a good understanding.

In recent years, a number of experimental investigations were conducted to evaluate the mechanical performance of fully grouted bolts subjected to tensile and shear loads. Bjurstrom [11] carried out shear tests on the bolts installed across a rock joint in laboratory and on site, and found that the host rock strength and the bolting angle (bolt axis with respect to joint surface) have substantial effects on the shear strength of the reinforced rock joint. Azuar [12] conducted shear tests on bolted concrete block specimens with smooth and undulating joints, and concluded that the joint friction features do not influence the shear resistance contribution of a bolt to a rock joint. Laboratory tests conducted by Haas [13] showed that an inclined bolt contributes more to the joint shear resistance than a perpendicular

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bolt and that pretension has no effect on the contribution. Dight [14] performed the shear tests with different bolting angles, and pointed out that a joint with a inclined bolt is stiffer than that with a perpendicular bolt and that the normal stress acting on the joint has no effect on the shear resistance of the bolt. However, experimental tests realized by Hibino and Motojima [15] revealed that the bolting angle has no clear influence on the shear resistance of the bolt.

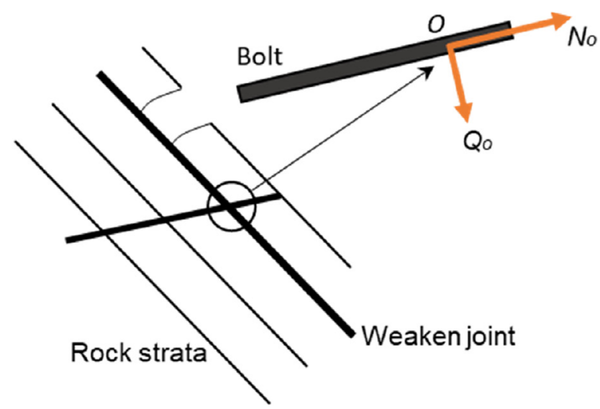
In addition, Ludvig [16], Spang and Egger [17] and Ferrero [18] continued laboratory shear tests on bolted rock joints with different inclined angles, host rock strengths and steel types, etc. Mchugh and Signer [19] discussed the bolt deformation behavior by using fourteen strain gauges installed in slots milled along the bolts. Aziz et al. [20], Jalalifar et al. [21], and Jalalifar and Aziz [22] evaluated the pretension effect on the performance of the grouted bolt by using a double-shearing apparatus and found that bolt contribution increases with increasing the bolt pretension. Grasselli [23] presented a series of shear tests on the samples, formed by three large concrete blocks with two symmetrical joints and bolted by two fully grouted bolts, and pointed out that the deformation behavior of the bolts can be divided into three distinct stages including elastic, non-linear and unconstrained plastic stages. Maiolino and Pellet [24] carried out shear tests on concrete specimens with large dimensions (1500 mm × 1000 mm × 625 mm, each) to investigate the influence of the bolt diameter on the shear strength of a bolted joint and concluded that the ultimate contribution of the bolt increases linearly with respect to the bolt diameter. Srivastava and Singh [25] examined the effect of the normal stress acting at the joint surface on the bolt contribution and reported that bolt contribution decreases with increasing the normal stress. Chen [26], and Chen and Li [27] studied the shear behavior of fully grouted bolts and reported that the greater the rock mass strength, the smaller the shear displacement. Although the present knowledge on fully grouted bolts enables us to interpret some phenomena of bolting effect in a qualitative way, the reinforced mechanism of the bolt as well as the evaluation of the internal forces mobilized in the bolt is still not thoroughly revealed. Besides, some of test results in the literatures contradict each other. For example, for the bolting angle effect, Bjurstrom [11], Azuar [12], Haas [13], Spang and Egger [17], Grasselli [23] and Maiolino and Pellet [24] pointed out that the bolting angle plays a significant role in effecting the bolt contribution, whereas Hibino and Motojima [15] and Chen and Li [27] claimed that the bolt contribution remains approximately constant no matter what the bolting angle is. Haas [13], Hibino and Motojima [15] and Ferrero [18] concluded that pretension reduces the shear displacement but does not influence the ultimate contribution of the bolt, while Jalalifar et al. [21] and Jalalifar and Aziz [22] reported that the higher the initial pretension load, the greater the contribution. Moreover, Spang and Egger [17] observed that the bolt contribution increases linearly with the section area of the bolt, on the contrary, Maiolino and Pellet [24] stated that the bolt contribution rises linearly with respect to the bolt diameter.

Therefore, there are clear needs for refinement of the understanding of the shear behavior of fully grouted bolts by the tests. In this work, direct shear tests were conducted on three groups of specimens with different bolting angles or grout strengths and shear loads, shear displacements and bolt strains were measured. The deflecting shape, the bending angle and the failure characteristic of the bolt were described in detail. The relationship of the shear load and the shear displacement was analyzed and the internal forces were determined according to the bolt strains or the shear displacement of the joint. Failure modes of fully grouted bolts subjected shearing was also discussed.

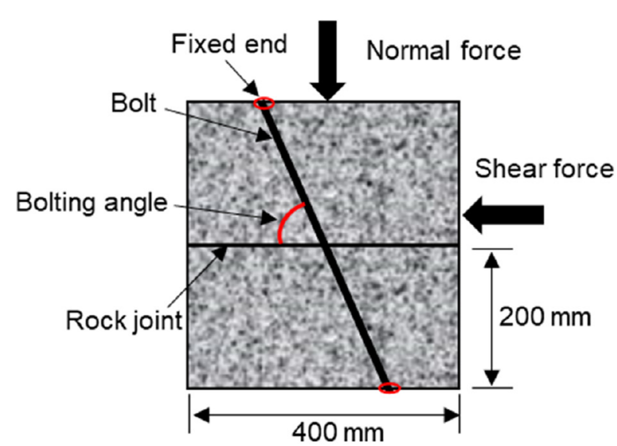
2. Testing program

2.1. Testing set-up

To simulate the traction and shear states of a fully grouted bolt in jointed rock mass, as a case shown in Fig. 1(a) where the bolt is obliquely set in bedding rock mass and both the axial load (N_0) and the shear load (Q_0) are activated in the bolt at the intersection of the bolt and the joint due to the sliding of the overlying rock mass along the joint, a testing model was proposed and depicted in Fig. 1 (b) in which concrete blocks were adopted to simulate the surrounding rocks and cement paste was used as grouting material. A direct shear test apparatus as depicted in Fig. 2 was developed to study the mechanical behavior of fully grouted bolts subjected to shearing. Two hydraulic cylinder systems can respectively apply shear and normal loads. The load capacities of two jacks applying normal and shear loads are respectively 100 kN and 200 kN. Four dial indicators were used to measure the shear displacements of the joints, and the normal displacements were recorded with other four dial indicators. Strains of the bolt during shearing were monitored by strain gauges. In order to reduce stress concentration at the joint surface due to the upper block twisting during shearing, the acting position of the shear load was located on the upper block as close to the joint plane as possible. Before the shear tests of the



(a)



(b)

Fig. 1. Testing model: (a) a bolting case in bedding rock mass. (b) Schematic diagram of shear test for the bolted rock joint.

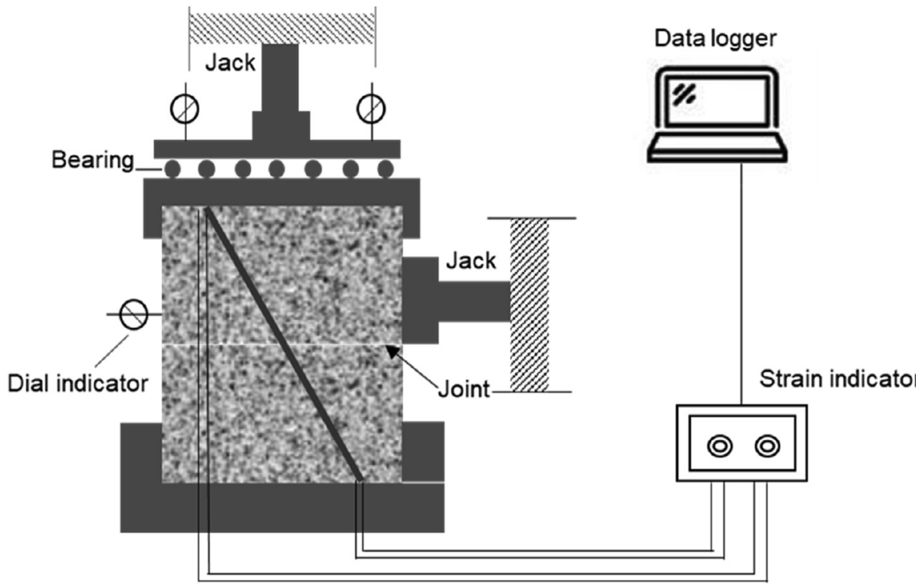


Fig. 2. Testing set up.

bolted rock joint were conducted, the unreinforced samples were sheared repeatedly to get a stable internal friction angle of the joint surface.

2.2. Sample preparation

Concrete is commonly used as an ideal model material to simulate the intact rock. Compared with natural rock masses, concrete blocks are easily fabricated, shaped and controlled to meet laboratory test requirements. In this study, testing samples consist of two concrete blocks and the size for each block is 400 mm × 400 mm × 200 mm. The lower block was firstly cast in a wooden mould with a PVC tube fixed through the center of the joint surface and intersecting with the joint plane at a given angle, as shown in Fig. 3. The external diameters of the PVC tubes for reserving grouting holes are 20 mm and 32 mm respectively corresponding to 8 mm and 12 mm of the bolt diameter. The inclined angles of the holes with respect to the joint plane were set as 90°, 75°, 60° and 45°, respectively. The upper block was cast after 24 h curing of the lower block. The blocks were cured for 28 days and then, sheared repeatedly for a stable friction angle of the joints.

The bolts are smooth round steel bars, of which the mechanical parameters were presented in Table 1. Nine pairs of strain gauges were affixed on the surface of the bolt with an interval of 15 mm

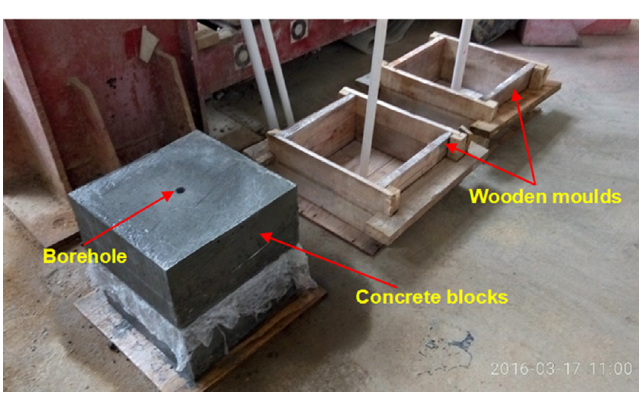


Fig. 3. Concrete blocks for testing.

Table 1
Mechanical parameters of the bolts.

Bolt diameter (mm)	Parameters
8	Young's modulus: 186 GPa Yield strength: 505 MPa Ultimate strength: 570 MPa Poisson's ratio: 0.28
12	Young modulus: 204 GPa Yield strength: 390 MPa Ultimate strength: 440 MPa Poison's ratio: 0.28

according to the industry recommended procedure and coated with 703 white silicone rubber to avoid being damaged during grouting and shearing process, as shown in Fig. 4. The bolt was inserted into the reserved borehole and fixed at the center of the hole, and subsequently, the hole was grouted with cement slurry or epoxy resin. Both ends of every bolt were fixed by the plates. Shear testing of the sample was carried out after another 28 days curing.

2.3. Testing procedure

Three groups of shear tests were performed as shown in Table 2. For group 1, bolts of 8 mm diameter were used with bolting angles of 90°, 75°, 60° and 45° for investigating the influence of the bolting angle on bolting behaviors. Four grout strengths of 29 MPa, 35 MPa, 48 MPa and 79 MPa, respectively corresponding to 0.5, 0.46 and 0.42 of the water-to-cement ratio of cement slurry and modified epoxy resin, were adopted to study the effect of the grout strength on the performance of the bolt for group 2. The Young's modulus of the grouts is 11.1 GPa, 11.5 GPa, 12.0 GPa and 6.5 GPa respectively with the grout strength increasing. The shear test of group 3 was conducted with nine pairs of strain gauges affixed on the bolt to reveal the deformation property of the bolt during shearing.

A given normal load was applied on the sample by the vertical jack, and before that, the dial indicators and strain gauges were reset. At the early stage of shearing, the shear load was applied with an interval of 0.5 kN by the horizontal jack. After the bolt

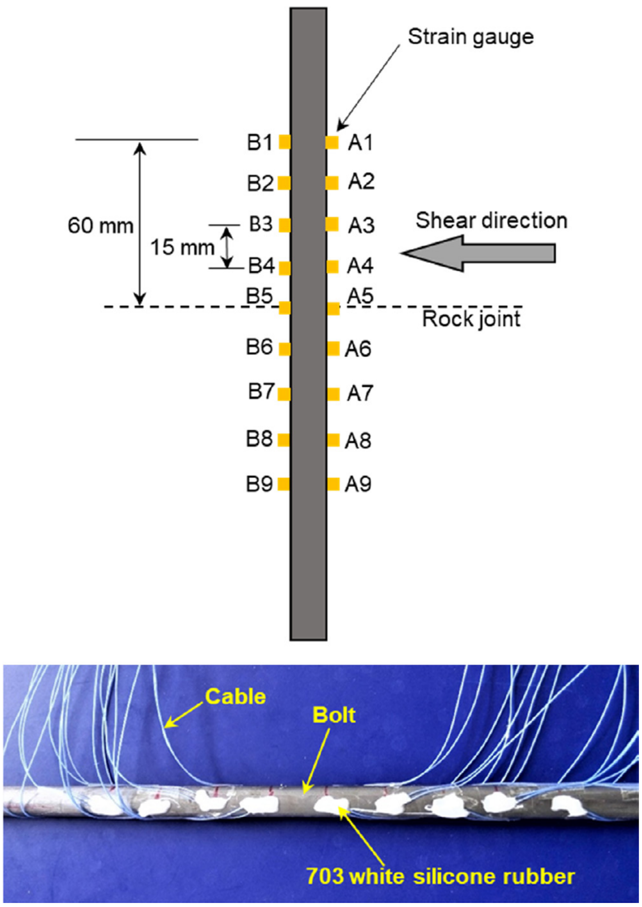


Fig. 4. Strain gauges for monitoring the deformation of the bolt during shearing.

reached the yielding state, the load step was controlled by shear displacement. Finally, failure of the bolt occurs once that the shear load decreased with increasing the displacement. At each loading step, the deformation monitoring data should be stable and recorded before next load was exerted. For observing failure characteristic of the bolt, the samples were sawed along the shear loading direction after tested.

3. Test results and discussions

Table 3 lists the shear tests carried out in the laboratory. The table gives the details of the testing parameters involved in this experimental program and the shear capacity of different bolt-joint specimens in shear tests. Failure descriptions of the bolts, relationships of shear displacement and shear load and bolting contributions as well as deflecting behavior of the bolts were presented and discussed.

3.1. Failure descriptions of the bolts

Fig. 5 shows the failure characteristic of group 1. It can clearly be seen from Fig. 5 that the bolt fails at the intersection of the bolt and the joint and that the bolt near to the joint deflects markedly, which indicates that the failure of the bolt is attributed to the combination of the axial and shear forces acting in the bolt rather than only the axial force or shear force. The deflecting deformation of the bolt is antisymmetric with respect to the joint and the transverse deformation length is about several times of the bolt diameter.

Fig. 6 shows the bending angles of the bolt (the bolt rotated angle at the failure respect to its initial axial direction) at the intersection of the bolt and the joint at different bolting angles. It can be found that the bending angles of the bolts with respect to their initial axial directions are 22°, 34°, 42° and 54° correspondingly to 45°, 60°, 75° and 90° of the bolting angle. The relationship of the bending angle and the bolting angle was depicted in Fig. 7. It is shown from Fig. 7 that the bending angle rises almost linearly with the bolting angle, which indicates that the shear force of the bolt plays a more important role while the bolt intersects with the joint at a greater angle. Besides, it is also observed that the greater the bolting angle, the longer the transverse deformation length of the bolt. Fig. 8 illustrates the influence of the grout strength on the bending angle, where the bending angles are 36°, 34°, 28° and 26° correspondingly to 29 MPa, 35 MPa, 48 MPa and 79 MPa of the grout strength. It is found that the bending angle is inversely proportional to the grout strength. Meanwhile, the transverse deformation length of the bolt decreases with ascending the grout strength.

Fig. 9 shows the angle between the fracture surface of the bolt and the joint plane. The angle measured in Fig. 9 is about 41°~42° while the bolting angle varies. This phenomenon suggests that the direction of the principal stress induced in the bolt steel at failure under shearing is not dependent on the initial bolting angle. In addition, it was found that the angle between the fracture surfaces of the broken bolt and the bolt bending direction is 65°, 67°, 75° and 77° corresponding with bolting angles range from 45° to 90°. The angle between the fracture surface of the broken bolts and the bolt bending direction, in some degree, reflects the failure mode of the bolts, which will be discussed in detail in the Section 4.

3.2. Relationships of shear displacement and shear load

During the shearing process of the bolted rock joint, the applied shear load and the associated shear displacement were recorded. The curves of the shear displacement and the shear load with different bolting angles and grout strengths were, respectively depicted in Figs. 10 and 11. It is clearly observed from Figs. 10 and 11 that the curves cover three stages including elastic, plastic hardening and failure stages. At the elastic stage, the shear load rapidly increases with a small shear displacement and their relationship is approximately linear. For the plastic hardening stage, it can be observed that, following a shorter yield phase, the shear load rises with increasing the shear displacement at a slower slope

Table 2
Testing groups.

	Group 1				Group 2				Group 3
Bolt diameter (mm)	8				8				12
Bolting angle (°)	90	75	60	45	60				90
Cement paste with water-to-cement ratio	0.44				0.50	0.46	0.42	Modified epoxy resin	0.44
Strain gauges	\					\			Nine pairs

Table 3
Test results.

Group No.	Sample No.	Bolt diameter (mm)	Bolting angle α (°)	Grout strength (MPa)	Concrete Strength (MPa)	Joint friction angle ϕ (°)	Normal load (kN)	Ultimate shear force (kN)	Bending angle of the bolt θ (°)
1	1	8	90	45.5	33	36.4	22	30.2	54
	2	8	75	45.5	33	42.5	22	46.5	42
	3	8	60	45.5	33	38.7	22	49.8	34
	4	8	45	45.5	33	42.1	23	53.3	22
2	5	8	60	29	33	44.7	40	74.8	36
	6	8	60	35	33	41.0	40	68.3	34
	7	8	60	48	33	42.3	30	62.8	28
	8	8	60	79	33	38.9	38	57.5	26
3	9	12	90	45.5	33	36.4	16	70.6	55

Notes: In this work, the bolting angle is the angle between the bolt axis and the joint plane. Joint friction angle is the inner friction angle of the joint plane. Bending angle of the bolt is the bolt rotated angle at failure respect to its initial axial direction.

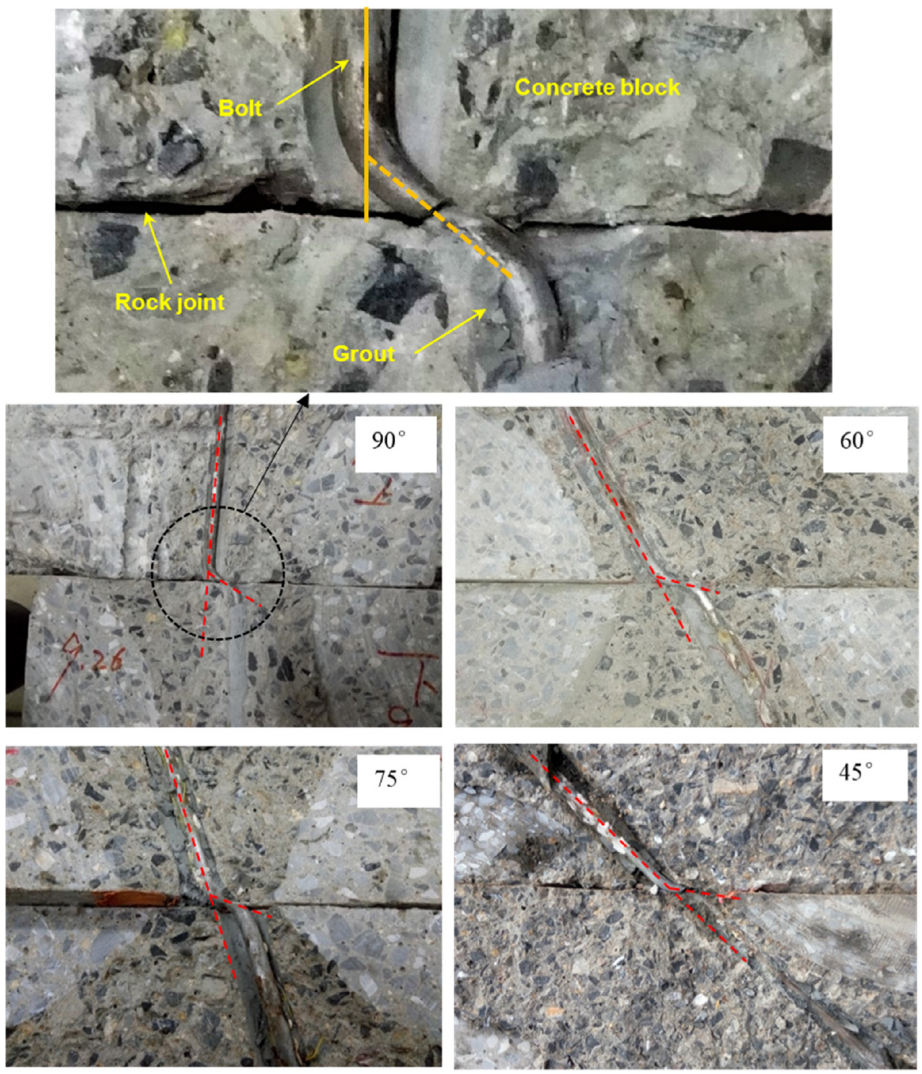


Fig. 5. Failure photos of the bolts with different bolting angles.

compared with the elastic stage. At the last stage, the shear load decreases sharply with ascending the shear displacement and the bolt fails accompanied by sounds.

As shown in Fig. 10, Within the range of 45° to 90° of the bolting angle, the maximum shear load of the specimens are 53.3 kN, 49.8 kN, 46.5 kN, and 30.2 kN respectively. The bolting angle has marked influence on the shear resistance of the bolted joint. Overall, the bigger the bolting angle, the smaller the shear resistance. Specially, the ultimate shear load at 45° of the bolting angle is 1.77 times of that at 90° of the bolting angle.

Fig. 11 shows that, in the elastic state, the shear stiffness of the bolted joint grouted by cement paste have no obvious difference for different water-to-cement ratios, but are larger than that grouted by modified epoxy resin, which is mainly due to the

Fig. 6. Bending angles of the bolt (the bolt rotated angle at the failure respect to its initial axial direction) at the intersection of the bolt and the joint with different bolting angles.

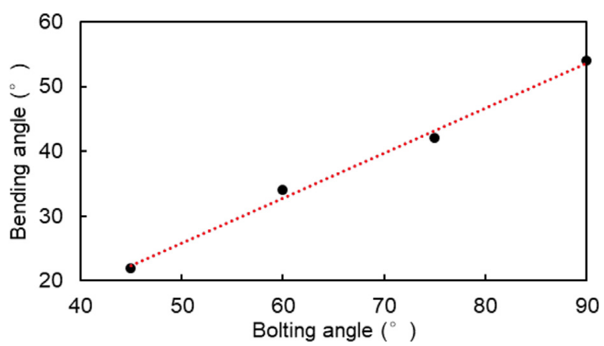


Fig. 7. Relationship between the bending angle (the bolt rotated angle at the failure respect to its initial axial direction) and the bolting angle.

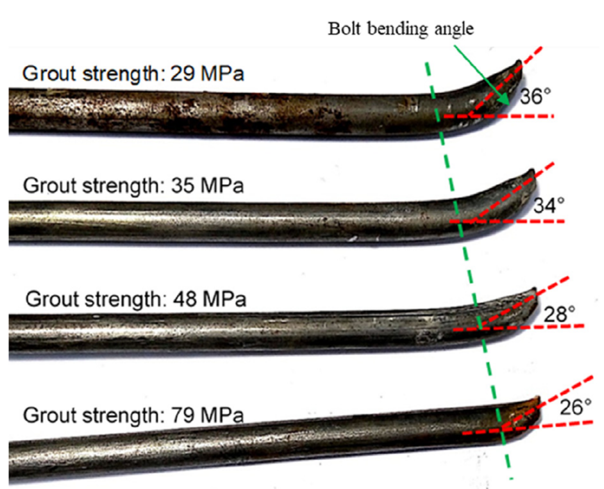


Fig. 8. Bending angles of the bolt at the intersection of the bolt and the joint with different grout strengths.

difference of the elasticity modulus between two grout materials. However, the maximum shear loads are respectively 74.8 kN, 68.3 kN, 62.8 kN and 57.5 kN for 29 MPa, 35 MPa, 48 MPa and 79 MPa of the grout strength, which means that, while other parameters keep unchanged, the larger of the grout strength, the smaller the shear resistance.

3.3. Bolting contribution

As frictional effect exists on the joint plane, it is complicated to directly interpret the bolt action. Bolt contribution is usually adopted for evaluating the increment of the shear resistance of the bolted joint [23]. The bolt contribution to the joint shear resistance, R, can be obtained by the following relationship:

$$R = T_S - T_N \tan\phi \tag{1}$$

where T_S and T_N are the applied shear load and the normal load, respectively, and ϕ is the joint friction angle.

By using the above expression, the curve of the bolt contribution against the bolting angle was obtained as shown in Fig. 12. With the bolting angle increasing, a marked decreasing tendency of the bolt contribution can be noticed. In other words, an inclined bolt provides a greater support capacity to restrain the joint shear movement than a perpendicular bolt. In addition, with increasing the bolting angle, the bolt contribution R follows the parabolic tendency as presented in Fig. 12, the bolting angle in horizontal axis which corresponding with the maximum values of the parabola is 48°, in other words, the bolt contribution is monotonically increasing when the bolting angle is varying between 0° and 48° and monotonically decreasing when the bolting angle ranges from 48° to 90°, and the optimal bolting angle of the bolt is about 48°. Therefore, to take advantage of the bolt potential, it is suggested to set the bolting angle moderate inclining with respect to the joint surface (40°~50°).

Fig. 13 shows the normalized bolt contribution standardized by the ultimate tensile load of the bolt, in which the normalized bolt contributions are 1.13, 1.09, 0.85, and 0.45 respectively corresponding to 45°, 60°, 75° and 90° of the bolting angle. It is clearly observed that, when the bolt steeply intersects the rock joint, the contribution of the bolt is less than its ultimate tensile load by a large margin. For example, the bolt contribution at 90° of the bolting angle is only 45% of the ultimate tensile load. As we know, the shear strength of the bolt is about half of its tensile strength, and the shear force (dowel effect) compared with the axial force plays a bigger role to the bolt contribution when the bolt angle is bigger. In other words, the larger the bolting angle, the greater the dowel effect of the bolt.

Fig. 14 depicts influence of the grout strength on the bolt contribution as the bolting angle is equal to 60°. A marked decrease tendency of the bolt contribution with increasing the grout strength can be seen from Fig. 14. The reason for this phenomenon is that a greater grout strength means a stronger restraint to the deflecting deformation of the bolt due to shearing and that, as a result, dowel effect of the bolt is stronger. It is also observed that the bolt contribution firstly decreases sharply with increasing the grout strength but, when the grout strength reaches a certain value (about 50 MPa in Fig. 14 for instance), the decrease is not very obvious.

3.4. Deflecting behavior of the bolts

Strain monitoring of group 3 was carried out to evaluate the deformation behavior of the bolts. Fig. 15 shows the strain variations of the bolt surface near the joint with increasing the shear displacement. In Fig. 15, there are nine strain gauges numbered A1–A9 in side A and B corresponding strain gauges numbered B1 to B9 in side B. Positive strain represents tensile strain while negative strain indicates compressive strain. Fig. 15(a)–(d) respectively show the strain distribution of the bolt near the joint when the shear displacements are equal to 2.86 mm, 3.91 mm, 4.56 mm and 5.23 mm. The followings can be observed from Fig. 15, as

- The corresponding point in side B would be in an opposite stress state while one point in side A is in traction or contraction, which reflects the deflecting feature of the bolt under shearing.

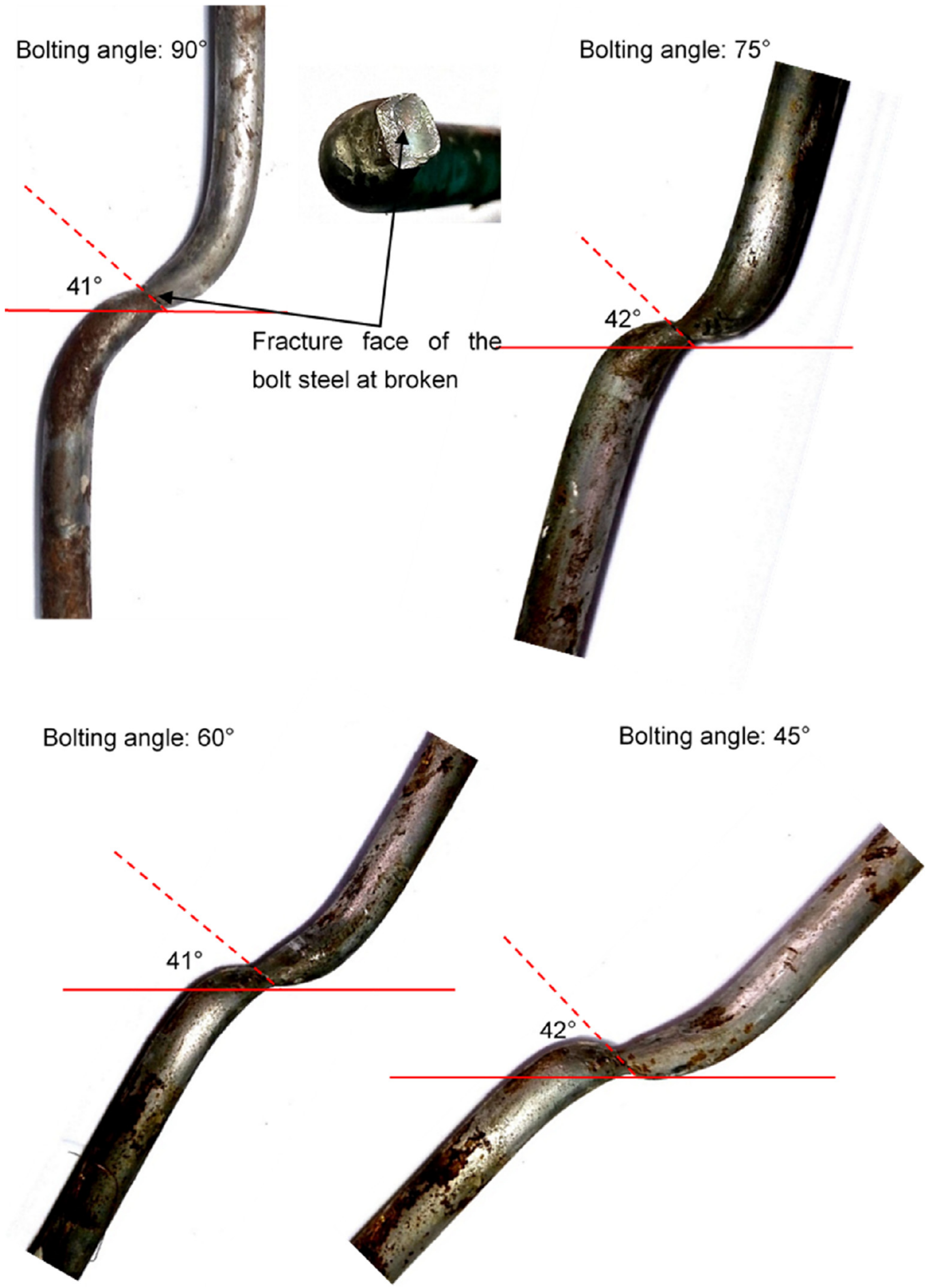


Fig. 9. Angles between the fracture surface of the bolt and the joint surface with different bolting angles.

- There exists a point of contraflexure of the bolt, which is identical to the theoretical model by Liu and Li [28], and several millimeters between the point of contraflexure and the intersection point of the bolt and the joint may be resulted from assembly error. The difference of the strains of a couple strain gauges indicates the amplitude of the bending moment. Two points of maximal bending moment occur respectively about 30 mm above and below the joint surface, and the distance between these two points is about 5 times of the bolt diameter (12 mm).
- While the shear displacement is small as shown in Fig. 15(a), absolute strain values of a couple strain gauges are approximately equal, which indicates that the shear force rather than the axial force of the bolt plays a dominant role. As the shear displacement increases (Fig. 15(b)–(d)), although the tensile strains grow faster than the compressive strains, the bending behavior caused by the bolt dowel effect still dominates.
- It can be judged from Fig. 15(c) and (d), where the bolt reaches the yield limit (the yielding strain of the bolt calculated from Table 1 is 1712 $\mu\varepsilon$), that the bearing capacity of the bolt still increases with increasing the shear displacement.
- Although it is possible for the two points of contraflexure of the bolt to develop into plastic hinge points, failure of the bolt occurs at the intersection point of the bolt and the joint rather than the points of contraflexure as shown in Fig. 5.

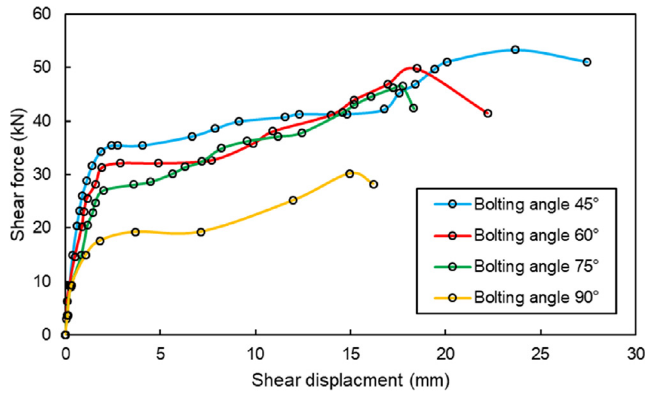


Fig. 10. Relationships of the shear displacement and the shear load with different bolting angles.

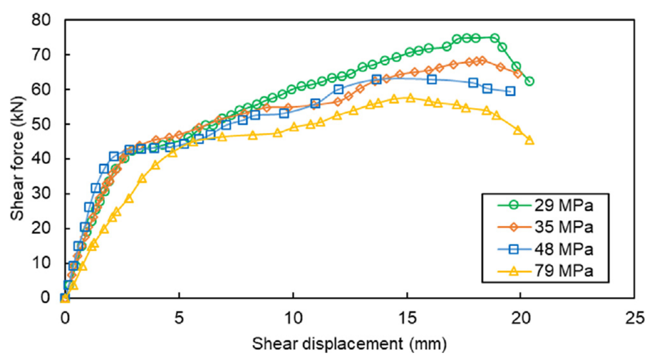


Fig. 11. Relationships of the shear displacement and the shear load with different grout strengths.

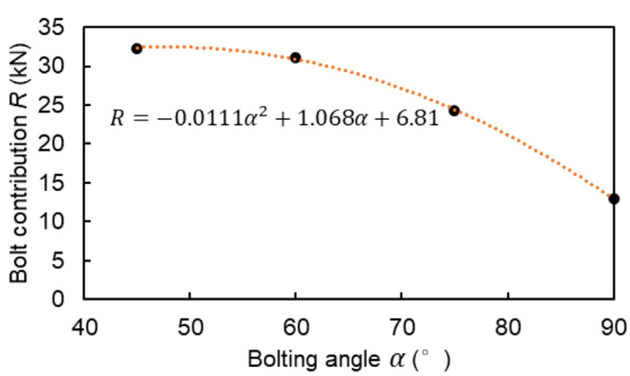


Fig. 12. Relationship between the bolt contribution and the bolting angle.

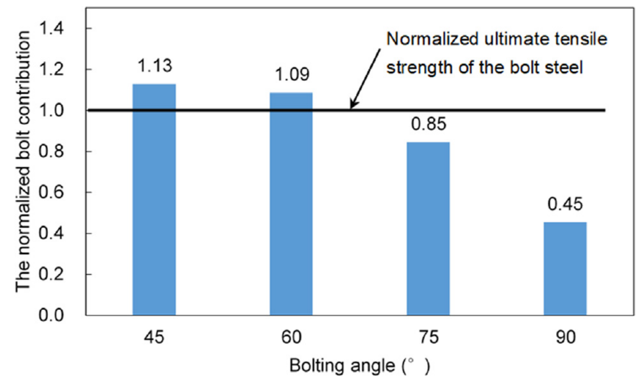


Fig. 13. Normalized bolt contribution standardized by the ultimate tensile load of the bolt versus the bolting angle.

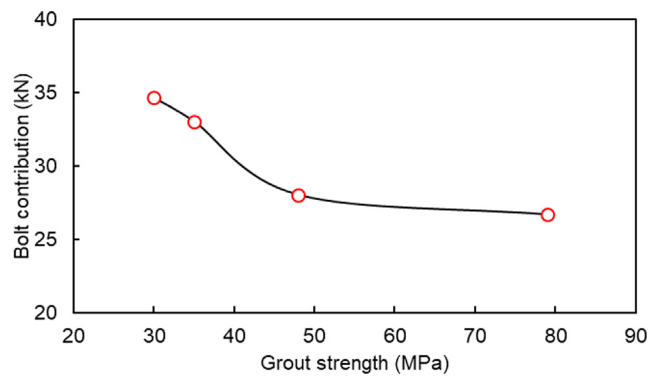


Fig. 14. Relationship between the bolt contribution and the grout strength with a bolting angle of 60°.

4. Analysis and discussions

4.1. Reinforcement mechanism of the rock bolt under shearing

As shown in Fig. 16, the bolt contribution to the shear resistance of a rock joint, R, provided by both the axial force and shear force, consists of two parts: one is the component parallel to the joint, R_c, providing additional cohesive of joint surface and here called as the cohesion enhancement effect. The other is the normal component to the joint, R_f, which increases the friction force of the joint surface and is defined as the friction enhancement effect. The values of R_c, R_f and R can be calculated by

$$R_c = N_o\cos(\alpha - \theta) + Q_o\sin(\alpha - \theta) \quad (2)$$

$$R_f = [N_o\sin(\alpha - \theta) - Q_o\cos(\alpha - \theta)]\tan\phi \quad (3)$$

$$R = R_c + R_f \quad (4)$$

where θ is the bending angle of the sheared bolt; α is the bolting angle, and ϕ is the joint friction angle.

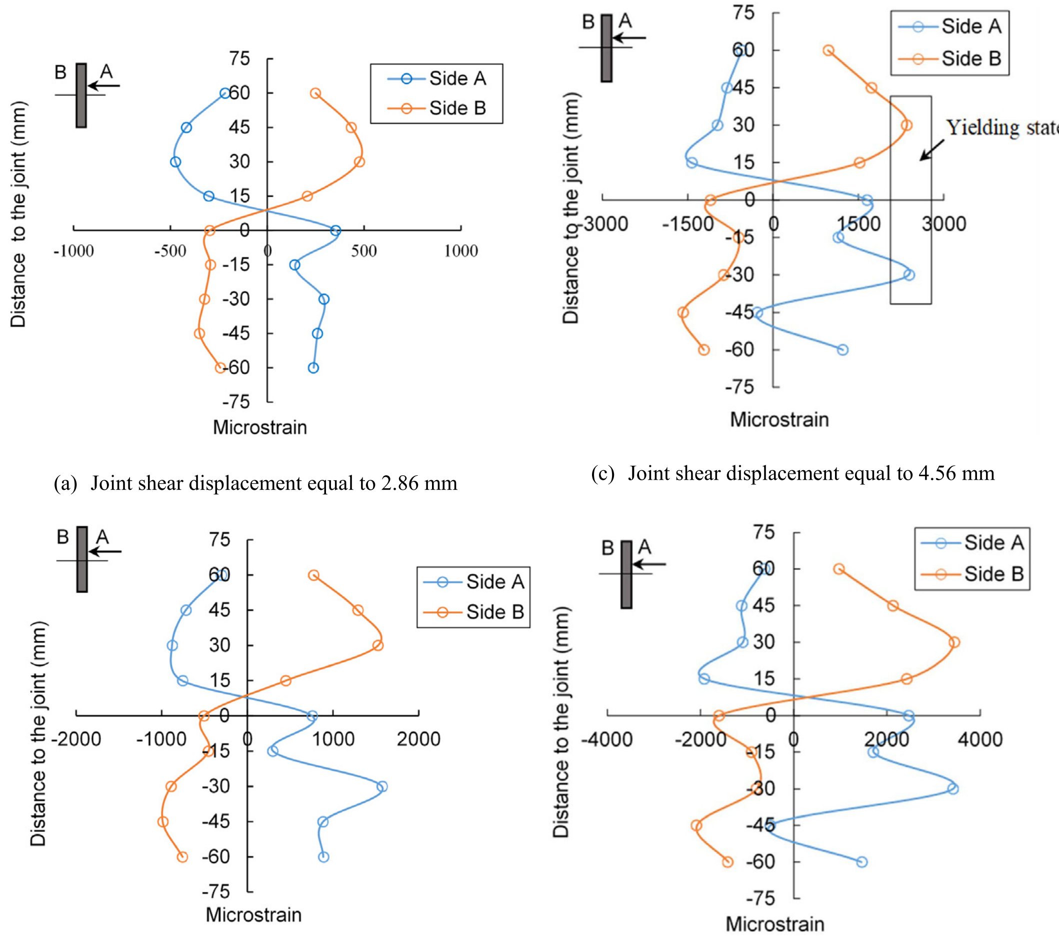
For the case of that failure occurs at the intersection point of the bolt and the joint, namely the point of contraflexure, the following failure criterion must be met [29], as:

$$\left(\frac{N_o}{N_f}\right)^2 + \left(\frac{Q_o}{Q_f}\right)^2 = 1 \quad (5)$$

where N_f and Q_f respectively represent the values of the axial and shear forces of the bolt at failure when they act separately and are equal to $A\sigma_f$ and $A\tau_f$, A is the cross section area of the bolt, and σ_f and τ_f are, respectively, the ultimate strengths of the bolt under pure tension and pure shear, with a form of $\tau_f = \frac{\sigma_f}{2}$ according to the Tresca's criterion.

As the bolt contribution R, bolt bending angleθ, joint friction angle ϕ and bolting angle α were all obtained by the shear tests or parameter testing, it is easy to calculate the axial force N_o and shear force Q_o as well as R_c and R_f using Eqs. (2)–(5). The results are presented in Table 4.

Figs. 17 and 18 respectively show the axial and shear forces acting at bolt-joint intersection with different bolting angles and grout strengths when failures occur. It is observed that, with increasing the bolting angles or grout strength, the axial force reduces while the shear force increases. The larger the bolting



(a) Joint shear displacement equal to 2.86 mm

(c) Joint shear displacement equal to 4.56 mm

(b) Joint shear displacement equal to 3.91 mm

(d) Joint shear displacement equal to 5.23 mm

Fig. 15. Strain distribution of the bolt near the joint with increasing the shear displacement.

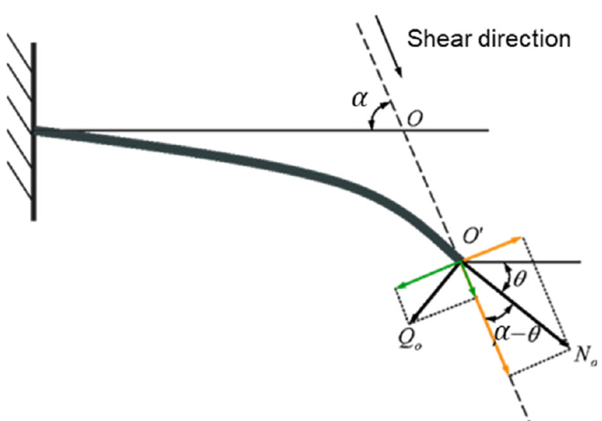


Fig. 16. Forces induced in the bolt.

Table 4
Values of the axial and shear forces in the bolt at the bolt-joint intersection at failure.

Test No.	Axial force N_o (kN)	Shear force Q_o (kN)	$\frac{Q_o}{N_o}$ (%)
1	11.1	13.2	118.9
2	20.0	10.2	51.0
3	25.8	6.2	24.0
4	27.0	4.8	17.7
5	27.7	3.7	13.3
6	27.1	4.7	17.4
7	22.7	8.8	38.6
8	21.7	9.4	43.6

angle, the greater the dowel effect. Particularly, when the bolt is installed vertically to the joint surface, the shear force is larger than that the axial force.

Figs. 19 and 20 respectively show the variations of R_c and R_f against the bolting angle and the grout strength. It can be observed that both R_c and R_f have a parabolic tendency as increasing the bolting angle or the grout strength. In the case of increasing the bolting angle, R_c and R_f increase first and then decrease, while they descend first and then rise for ascending the grout strength. It is also found from Figs. 19 and 20 that R_c is always much larger than R_f, which is mainly contributed to that the shear force, Q_o, directly enhances the shear resistance of the joint but decreases the normal pressure of the joint as presented in Eqs. (2) and (3). Fig. 19 show that, when the bolt steeply intersects the joint (e.g., the bolting angle is more than 80°), R_f is negative, which indicates that dowel effect can reduce the bolt contribution by a large margin as depicted in Figs. 12 and 13.

4.2. Failure modes of the bolts under shearing

As observed from the above shear test results, the broken of the bolts are all located at the bolt-joint intersection, regardless of different bolting angles or grout strengths. Fig. 21 shows the stress state of the fracture surface of the bolts. According to the angle

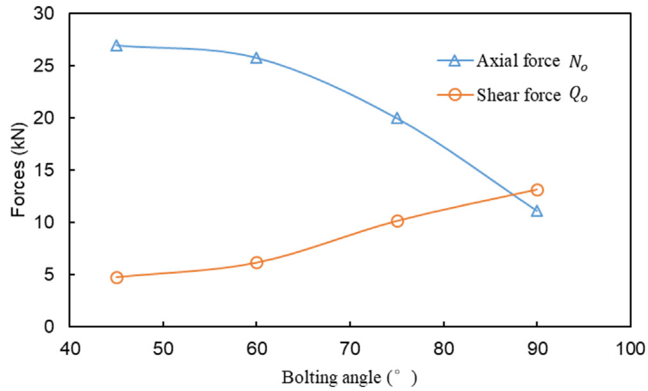


Fig. 17. Axial and shear forces acting at bolt-joint intersection with different bolting angles.

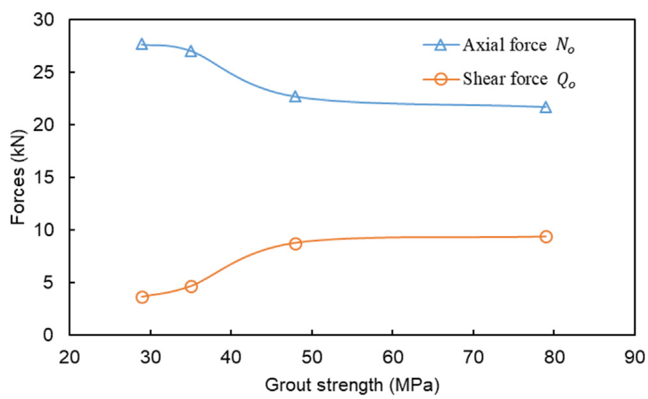


Fig. 18. Axial and shear forces acting at bolt-joint intersection with different grout strengths.

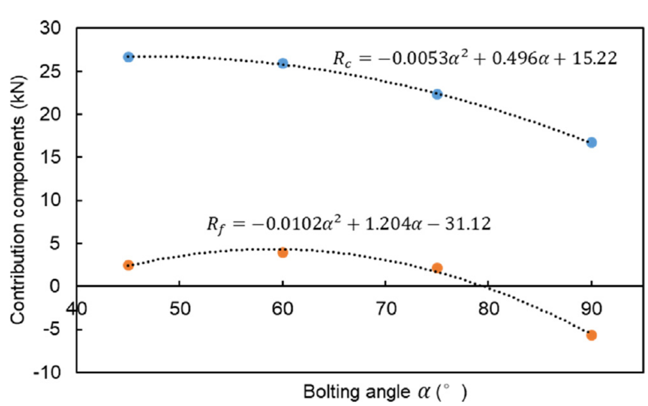


Fig. 19. Contribution components R_c and R_f against the bolting angle.

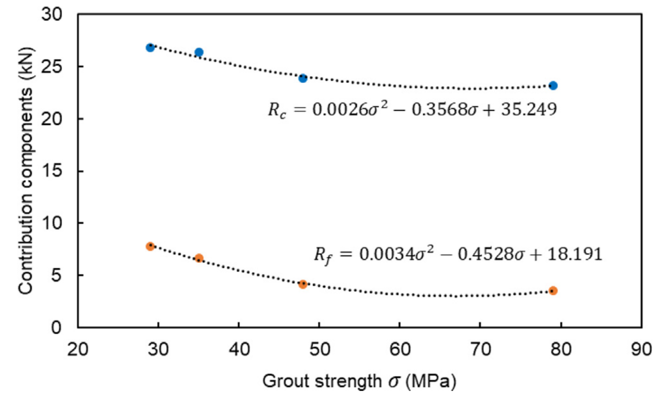


Fig. 20. Contribution components R_c and R_f against the grout strength.

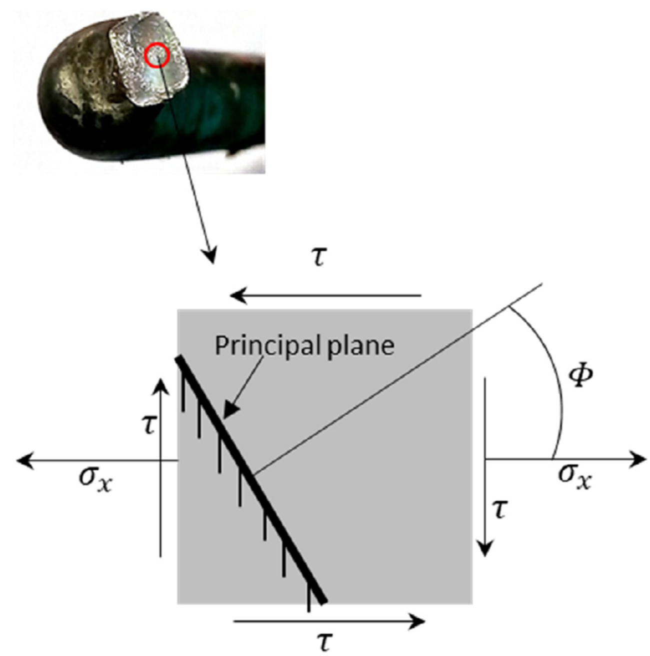


Fig. 21. Stress state of the broken surface of the bolt, σ_x is axial stress and τ is shear stress.

between the fracture surface of the bolt and the bolt bending direction obtained in Section 3.1, Φ (the angle between the bolt bending direction and the normal to the principal plane) are 25°, 23°, 15°, and 13° respectively corresponding with the bolting angles ranges from 45° to 90°. For cases when the failure of the bolts caused by pure tension and pure shear, as shown in Fig. 22, Φ are 0° and 45° respectively. This indicates that the failure of the bolts subjected to shearing are not caused by the separate action of axial force or shear force, but by the joint action of them two.

Ferrero [18] proposed two failure modes of bolts. It was considered that the failure mechanism of bolts dues to the axial force after the formation of two hinges points with respect to the shear plane when the strength of surrounding rock is less than 50 MPa, and the failure mode of bolts are combination of axial and shear forces when the strength of the surrounding rock greater than 50 MPa. However, in our shear tests, regarding the strength of the surrounding rock mass is 33 MPa, and the failure of the sheared bolts were all caused by the joint action of axial and shear forces.

4.3. The relation of the axial force and shear force during shearing

For a bolt subjected to shearing, the relationship between the axial force N_o and shear force Q_o acting in the bolt at the bolt-joint intersection is essential to establish the anchorage model of the reinforced system and to determine the bolt contribution. Based on testing results, as discussed above, Eqs. (2)–(5) can quantitatively determine the axial force N_o and shear force Q_o at failure but fail to provide the values of the forces under arbitrary loads or shear displacements. By using the strain data of group 3 presented in Section 3.4, the axial force at different shear displacements can be obtained according to Hooke's law.

In the shear test for group 3, nine points (a pair of strain gauges attached at side A and side B of each point as depicted in Fig. 4)

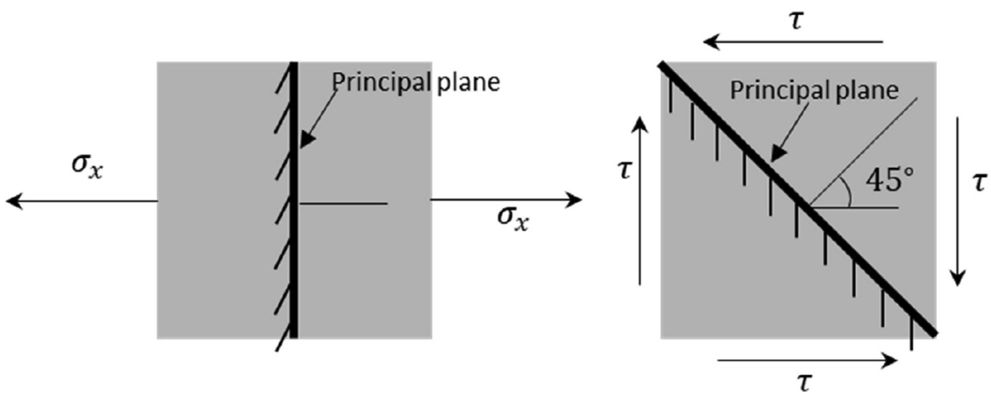


Fig. 22. Stress state of the bolt in pure tension and pure shear, respectively.

were chosen to monitoring the strains of the bolt due to shearing. To illustrate the relationship of the axial force and the strain of the bolt, point 5 with strain gauges A5 and B5, which was planned to locate at the intersection point of the bolt and the joint and was in fact found to be several millimeters away from the intersection point was adopted to be analyzed. The tension model of fully grouted bolts proposed by Li and Stillborg [3] indicates that the difference in axial force due to the installation error is very small and can be neglected. Besides, it should be pointed out that the strains in the elastic stage rather than in the plastic stage were recorded due to that all gauges were damaged in plastic stage. According to beam theory, the axial stresses of point 5 on sides A and B can be expressed respectively:

$$\sigma_A = \frac{N}{A} + \frac{M}{W} = E \cdot \varepsilon_A \tag{6}$$

$$\sigma_B = \frac{N}{A} - \frac{M}{W} = E \cdot \varepsilon_B \tag{7}$$

Combining Eqs. (6) and (7) yields:

$$N = \frac{EA \cdot (\varepsilon_A + \varepsilon_B)}{2} \tag{8}$$

where E is the Young's modulus of the bolt steel, ε_A and ε_B are respectively the strains of A5 and B5.

Based on Eq. (8), the axial force at point 5 with different shear displacements was calculated and presented in Fig. 23, from which an exponential growth tendency of the axial force was observed.

For the shear force Q_o, it is difficult to measure directly in the shear test. A feasible solution to obtain the shear force is to use

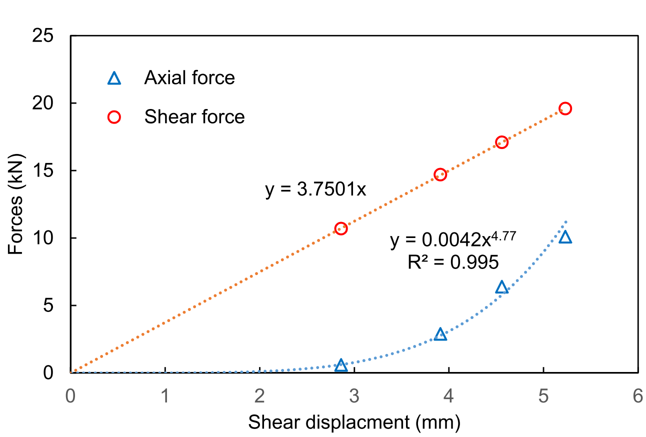


Fig. 23. Axial and shear forces acting at bolt-joint intersection obtained by the shear test.

the relation of the shear force and the deflecting deformation. Di Prisco [30] and Pellet and Egger [31] showed that the deflection deformation of the bolt in the elastic stage can be represented by the following function

$$v(x) = v_o e^{-x/l} \cos\left(\frac{x}{l_o}\right) \tag{9}$$

where v_o is the joint shear displacement; l_o is a reference length of the rock bolt, and $l_o = \sqrt[4]{\frac{4EI}{kD}}$, I is the moment of inertia, D is the rock bolt diameter, and k is the Young's modulus of the grout.

The shear force Q_o can be deduced based on the differential relationships among the deflecting deformation and internal forces, as:

$$Q_o = \frac{2EI v_o}{l_o^3} \tag{10}$$

The relationship of the shear force Q_o calculated by Eq. (10) and the shear placement of the rock joint was depicted in Fig. 23. It is found in Fig. 23 that the shear force rises linearly with increasing the shear displacement of the joint, and that the axial force is clearly lower than the shear force although it grows rapidly.

Combing the equations represented respectively by the shear force and axial force in Fig. 23, the relationship between the shear force Q_o and axial force N_o can be derived:

$$N_o = 0.0042\left(\frac{Q_o}{3.7501}\right)^{4.77} \tag{11}$$

where the unit of N_o and Q_o are kilo Newton.

Clearly, the relationship between the axial force and shear force obtained above is not a universally applicable formula, on the one hand, only limited tests undertaken, and on the other hand, the strain behavior of the bolt in plastic hardening and failure stages was not properly captured due to the strain gauges attached on the bolt surface were easily damaged. Besides, taking the value of the axial force at point 5 as that at the bolt-joint intersection may bring about errors.

5. Conclusions

Experimental investigations have been carried out to study the shear behavior of fully grouted bolts with different bolting angles or grout strengths. The main observations and conclusions can be drawn as follows.

It is clearly observed that the bolt near the rock joint bends drastically at failure and the deflecting deformation of the bolt is antisymmetric with respect to the joint, which indicates that the failure of the bolt is attributed to the combination of the axial and shear forces acting in the bolt rather than the axial force or shear force acts separately.

The bending angle and the transverse deformation length of the bolt rise with increasing the bolting angle but decrease with ascending the grout strength. However, the angle between the fracture surface of the bolt and the joint plane almost keeps constant while the bolting angle varies.

Due to the deflection of the bolt under shearing, the cross section of the bolt contains a tensile zone and a compressive zone. In the elastic stage, the axial force acting in the bolt can be calculated according to the different value of the maximal tensile stress and the maximal compressive stress of the cross section which can be measured by strain gauges. In the case of the bolt steeply intersecting the joint, the axial force at the intersection of the bolt and the joint is very small in the initial shear stage and then increases with the shear displacement in an exponential form. Besides, it was found that two points of maximal bending moment formed in the bolt and about several times of the bolt diameter away from the joint surface.

Although two failure modes including tensile-shear failure at the bolt-joint intersection and tensile failure between two maximal bending moment points were proposed in literature and 50 MPa of the uniaxial compressive strength of rock was recommended by Ferrero [18] as a threshold value for the mode transition from tensile failure between two plastic hinges points to tensile-shear failure, the test results reveal that, in the case of the rock strength equal to 33 MPa, all bolts fail at the bolt-joint intersection in a tensile-shear mode.

With increasing the bolting angle, the bolt contribution rises first and then reduces, and they obey a parabola relationship. To take advantage of the bolt potential, therefore, it is suggested to set the bolting angle moderate inclining with respect to the joint surface (bolting angle $40^\circ \sim 50^\circ$).

A marked decrease tendency of the bolt contribution with increasing the grout strength. Specifically, the bolt contribution firstly decreases sharply with increasing the grout strength but, when the grout strength reaches a certain value (about 50 MPa, for the case in Fig. 14), the decrease is not very obvious.

Declaration of Competing Interest

None.

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